2. Installing ROS 2 Jazzy

- ROS 2 Jazzy corresponds to Ubuntu 24.04. This tutorial uses Ubuntu 24.04 to install ROS 2 Jazzy.
- The software sources, packages, and interfaces of different Ubuntu and ROS 2 versions may differ. Use the ROS 2 version that matches your system; do not simply replace the version name and run the commands directly.
- The factory system already comes with the Jazzy ROS environment configured. This tutorial applies to a clean system.

1. Set the Locale

First check whether the local locale supports UTF-8 encoding. You can run the following command to check and set the UTF-8 encoding.

```
locale  # Check whether UTF-8 is supported

sudo apt update && sudo apt install locales
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8

locale  # Verify that the setting is successful
```

Note: The locale can be different, but it must support UTF-8 encoding.

2. Set the Software Source

Start the Ubuntu universe repository.

```
sudo apt install software-properties-common
sudo add-apt-repository universe
```

Install the ROS 2 software source configuration package. This method automatically configures the GPG key and the software source matching the current Ubuntu version:

```
sudo apt update && sudo apt install curl -y
export ROS_APT_SOURCE_VERSION=$(curl -s https://api.github.com/repos/ros-
infrastructure/ros-apt-source/releases/latest | grep -F 'tag_name' | awk -F\"
'{print $4}')
curl -L -o /tmp/ros2-apt-source.deb "https://github.com/ros-infrastructure/ros-
apt-source/releases/download/${ROS_APT_SOURCE_VERSION}/ros2-apt-
source_${ROS_APT_SOURCE_VERSION}.$(. /etc/os-release && echo
${VERSION_CODENAME})_all.deb"
sudo dpkg -i /tmp/ros2-apt-source.deb
rm -f /tmp/ros2-apt-source.deb
```

3. Install Jazzy

First, update the apt repository cache:

```
sudo apt update
```

Then upgrade the installed software (ROS 2 packages are built on the frequently updated Ubuntu system; make sure your system is up to date before installing new packages):

```
sudo apt upgrade
```

Install the desktop version of ROS 2 (recommended), which includes ROS, RViz2, examples, and tutorials:

```
sudo apt install ros-jazzy-desktop python3-argcomplete
```

Install the colcon build tool.

```
sudo apt install python3-colcon-common-extensions
```

4. Configure the Environment

In a terminal, before running ROS 2 programs, you need to run the following command to configure the environment:

```
source /opt/ros/jazzy/setup.bash
```

You have to run the above command every time you open a new terminal. Alternatively, you can run the following command to write the environment configuration command into the "~/.bashrc" file, so that you do not need to configure the environment manually each time you start a new terminal.

```
echo '[ -f /opt/ros/jazzy/setup.bash ] && source /opt/ros/jazzy/setup.bash' >>
~/.bashrc
```

After reopening the terminal, use the following commands to check the environment:

```
echo $ROS_DISTRO
command -v ros2
```

You should see `jazzy` and `/opt/ros/jazzy/bin/ros2` respectively. If you also set `PYTHONPATH`, do not overwrite the ROS 2 Python package search path; appending is recommended, for example:

```
export PYTHONPATH="/home/jetson/jetcam${PYTHONPATH:+:$PYTHONPATH}"
```

At this point, ROS 2 has been installed and configured.

5. Run the Publisher and Subscriber Example Nodes

The ROS 2 desktop version has integrated some example nodes. You can test whether ROS 2 is installed successfully by running the example nodes.

- Run the publisher node

```
ros2 run demo_nodes_cpp talker
```

- Run the subscriber node

```
ros2 run demo_nodes_cpp listener
```

6. Run and Experience the Turtle Case

The turtle simulator turtlesim is a classic teaching tool in ROS. Through turtlesim, you can quickly understand the core concepts in ROS.

- Run the turtle simulator

```
ros2 run turtlesim turtlesim_node
```

- Run the keyboard control node

```
ros2 run turtlesim turtle_teleop_key
```

- View the velocity topic data

```
ros2 topic echo /turtle1/cmd_vel
```